FLEET TEAM DIGEST

737MAX-FTD-24-22003

Issue Title : Power Distribution Panel (PDP) P91/P92 Contactor Failures - Generator Control Breaker (GCB) / Auxiliary Power Breaker (APB) / Bus Tie Breaker (BTB)

Airplane Model	ATA	Minor Model(s)
737MAX	2421-41	-7, -8, -8200, -9, -BBJ
Other Model(s)	**ECCN**	
	9E991	

Originated Date	Last Revised Date	Created On
12/21/2022	11/08/2023	08/14/2023
Estimated Completion Date	**Next Update Date**	

Revision Description

To close the FTD.

Description

Operators have recently reported power transfer related issues in the MAX fleet, which have been linked to Generator Control Breaker (GCB), Auxiliary Power Breaker (APB) or Bus Tie Breakers (BTB) contactors failures, as follows:

- stuck open
- stuck closed
- uncommandable state (meaning a trip signal would not open the main contacts or a closed signal would not close the main contacts)
- aux contacts not reflecting the main contactor state
- "sluggish" main contacts

There have been recent reports of multiple operator and production removals.

As a result of an APB stuck closed condition, some operators have experienced an issue with not being able to apply AC Power (APU Power or External Power) to the aircraft when the main engines are shutdown. This is because depending on the stuck condition of the relay contacts (Main and/or Aux) the relay can be in a state where the APB is being sensed in a closed position. If the APB is being sensed in a closed position the following aircraft effects may happen:

1. Due to internal Bus Power Control Unit (BPCU) logic, the External Power Contactor (EPC) will not receive a ground signal to allow the External Power on the aircraft, regardless of the connected External Power is within the power specifications to bring power onto the aircraft.
2. Internal BPCU logic will also not allow the Ground Service Bus to be energized.
3. The APU start converter unit monitors the position of the auxiliary contacts in the auxiliary power breaker (APB) for the

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/04/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

airplane and compares this position with its internal isolation relay. If the auxiliary contacts show that the auxiliary power breaker or the isolation relay in the start converter unit is in the incorrect position for the APU start mode, then this fault "49-41247 START CONVERTER UNIT SHOWS INTERLOCK RLY FAIL" is set. The start converter unit will prevent an APU start and the fault signal goes to the Electronic Control Unit (ECU).

4. Without being able to bring APU Power, External Power or energize the Ground Service Bus onto the aircraft, with the Main Battery Switch set to "On", the battery will be the only source of power available until the APB is replaced.

An investigation was initiated to determine the cause of contactor failures in service which included Boeing on-wing testing and TE Connectivity engineering investigation. The Investigation revealed that some failures resulted in a stuck open, stuck closed or an uncommandable state (meaning a trip signal would not open the main contacts or a closed signal would not close the main contacts). The investigation also revealed that the contacts could fail such that the main contacts and the aux contacts could be in disagreement with the contactor state.

During the investigation, one of the problems that occurred was a "sluggish" closing of TE Connectivity main contacts. This condition resulted in a lack of inertia present at the time the auxiliary switches were initially adjusted to actuate the microswitches and therefore could not properly change state. As a remedy, TE Connectivity implemented optimizations to the microswitch adjustment process to help improve performance (aux contact actuation). After adjustment, the "stale-mate" or uncommandable condition was not able to be repeated. This means, regardless of the current present at the time of close, if the main contacts changed state fully, so will the aux contacts.

The investigation resulted in a contactor switch setting adjustment being implemented (S/N cut-in CW-13769). Since the contactor micro switch adjustment implementation at TE Connectivity, Boeing has not observed any failed contactors in-service.

Status

Boeing, Honeywell, Celestica and TE Connectivity are engaged and monitoring the fleet removals.

This contactor issue was reported through the Continued Operational Safety Program (COSP) reporting process at Boeing and determined to be Not Safety.

Interim Action

If operators experience a similar power transfer related issue, Boeing recommends replacing the failed contactor P/N 51080205-001 with a contactor post S/N CW-13768, a contactor pre S/N CW-13769 that has been reworked per the new TE Connectivity micro switch adjustment process, or contactor P/N 1151968-1. Contactor P/N 51080205-001 S/N numbering system is in an alphabetical order. For example, serial numbers that start with CN, CU and CV are pre S/N CW-13769 regardless of the numerical digits.

Contactor P/N 1151968-1 is used only on in-service aircraft but is no longer delivered on production aircraft as this P/N was

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/04/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

replaced by 51080205-001 per PRR 3M0057.

Boeing and Honeywell have changed the related P91 and P92 CMM and AIPC to allow use of the previous contactor P/N 1151968-1 used on the 737NG aircraft. Both 51080205-001 and 1151968-1 are fully interchangeable on the MAX aircraft.

Final Action

SIL 24-C-179 is released. Upon unit shop visit, review the SIL for applicability to the removed Contactor part # 51080205-1 and incorporate the change in the SIL if applicable. Please report to Boeing any failures/removals of contactors during power switching. Report P/N, SN and Time Since New (TSN); PDP P/N, S/N and TSN; APB, GCB or BTB location. Operators are recommended to monitor contactor performance.

Milestones

Max Grounding - 03/2019
Customer Initial Complaint Submission to TE Connectivity on Contactor Failures - 03/2020
TE Connectivity Engineering investigation and analysis - 03/2020
TE Connectivity Contactor build and adjustments process change solution - 06/2020
Boeing P91/P92 CMM and AIPC Change - 07/15/2021

Contactors prior to S/N CW-13769 that have been reworked and adjusted per the new micro switch adjustment process will be marked with blue vinyl dot that is covered with a permanent acrylic adhesive and a 1 mm thick, clear, FASTCAL 700 polyester cover. Honeywell issued Service Information Letter (SIL) No 24-C-179 on the retrofit information.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Component Maintenance Manual (CMM)	24-50-61		
Component Maintenance Manual (CMM)	24-50-62		

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
GCB/APB/BTB		TE Connectivity	KAL290KS (51080205-001).
GCB/APB/BTB		ECE/Zodiac	1151968-1
PDP		Honeywell	51090384-00XX and 10XX

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/04/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991

PDP		Honeywell	51090386-00XX and 20XX

Related Categories

- Component Issues : Issues that occur during component maintenance or have an impact on component reliability
- Dispatch Reliability Issues : Issues that cause revenue flight delays and cancellations

BOEING PROPRIETARY - The information contained herein is Proprietary to The Boeing Company and shall not be reproduced or disclosed in whole or in part or used for any reason except when such user possesses direct, written authorization from The Boeing Company. Encrypted emails required for forwarding via Email.The document generated on 11/04/2026 by Fleet Team Digest application via MyBoeingFleet. This document is for reference purposes only, visit the application for all the latest and updated revisions of this document.

Copyright ©2026 Boeing. All rights reserved.

EXPORT CONTROLLED -This technology or software is subject to the U.S. Export Administration Regulations (EAR), (15 C.F.R. Parts 730-774). No authorization from the U.S. Department of Commerce is required for export, re-export, in-country transfer, or access EXCEPT to country group E:1 or E:2 countries/persons per Supp.1 to Part 740 of the EAR. ECCN : 9E991